

RITIK GUPTA

+91-8923937302 Porwal2000ritik@gmail.com
linkedin/ritikgupta-rp github/Ritik-Gupta8 ritikgupta8.in

Summary

Full-stack and applied AI engineer building end-to-end, production-grade LLM-powered web applications. Ships fast, iterates faster. From a bilingual AI crop-disease app deployed on AWS to a multi-agent quiz platform with an 8-model fallback chain. Comfortable across the full stack: Python backends, React/Vue frontends, cloud infra, and generative AI integration.

Education

Indian Institute of Technology Madras	Sep 2023 – Nov 2025
B.S. in Data Science and Applications	Chennai, India
Chhatrapati Sahuji Maharaj University	Aug 2019 – Aug 2022
B.Sc. in Mathematics, Physics & Economics	Kanpur, India

Projects

KrishiRakshak — AI Crop & Livestock Doctor | Flask, Gemini 2.5 Flash, AWS S3, DynamoDB, EC2, PWA2026

- Situation:** Indian smallholder farmers lose 20–40% of crops annually due to delayed or unavailable disease diagnosis.
- Task:** Built a bilingual (Hindi/Marathi) multimodal disease-detection app for the **AWS AI for Bharat Hackathon**, targeting low-connectivity rural users.
- Action:** Engineered a vision-language pipeline using Gemini 2.5 Flash with two-stage prompt chaining — image validation then structured JSON output (disease, severity 0–100, treatment, prevention); integrated OpenWeather API for humidity/temperature-based risk alerts; architected AWS backend (S3 + DynamoDB) with client-side image compression for offline-capable delivery.
- Result:** Delivered a live, EC2-deployed prototype with a Chart.js analytics dashboard tracking diagnosis history and severity trends; accepted as a qualified hackathon submission. [GitHub](#)

Quiz Master — AI-Powered Adaptive Quiz Platform | Flask, PostgreSQL, Supabase, Gemini API, Tailwind, Render2026

- Situation:** Generic quiz tools lack adaptive AI generation, cost reliability, and concurrent multi-user isolation — problems that become critical at scale.
- Task:** Build a production-grade, multi-role quiz platform with zero-cost AI generation and robust session handling.
- Action:** Designed an 8-model free-tier fallback chain across Gemini variants to ensure 99%+ AI uptime at \$0 cost; implemented SHA-256 SQL response caching to eliminate redundant LLM calls; enforced anti-cache-poisoning headers and session scoping for data isolation across concurrent users; built XP-based leaderboard and real-time Chart.js analytics dashboards.
- Result:** Platform live on Render with Supabase-hosted PostgreSQL, supporting multi-user concurrent sessions; caching measurably reduced repeat-query latency. [Demo](#) | [GitHub](#)

Smart Parking Management System | Flask, Vue.js, Redis, Celery, SQLite, Bootstrap, Render2025

- Built a full-stack role-based parking management web app with reservation workflows, dynamic pricing, caching, background jobs, and administrative dashboards.
- Designed, versioned, and stress-tested 15+ REST API endpoints in Postman, enforcing schema validation and edge-case guards across reservation, billing, and release workflows. [Demo](#) | [GitHub](#)

Technical Skills

Languages & Frameworks: Python, JavaScript (ES6+), SQL, Flask, React.js, Vue.js, Tailwind CSS
AI & LLM: Google Gemini API, Prompt Engineering, Prompt Chaining, Multi-Agent Systems, Gemini ADK
Cloud & Databases: AWS (EC2, S3, DynamoDB), Google Cloud, PostgreSQL, Supabase, Redis, SQLite
DevOps & Tools: Docker, GitHub Actions, CI/CD, Celery, Postman, Git, Vercel, Render, Linux/WSL

Certifications & Experience

Google Cloud Career Launchpad: Generative AI Leader — Google CloudJan–Apr 2026

Foundation models, RAG, AI agents, responsible AI on GCP. [View](#)

Full-Stack Web Development Internship — Jyesta Corporate EntitySep–Nov 2025

Built and shipped backend features: REST API design, database schema, and third-party API integration. [View](#)

AWS Academy Graduate: Cloud Foundations — AWSMay 2026

[View](#)